

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1596

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Code of Conduct

I Bhuvaneshwari J (192572141)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.

Experiment 2: Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 3 CPU, 10GB RAM and 10GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

Assessment Tasks

Experiment 1: Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as name of the student, Reg. no. of the student, Subjects, Marks, Average, percentage, rank, overall performance etc.

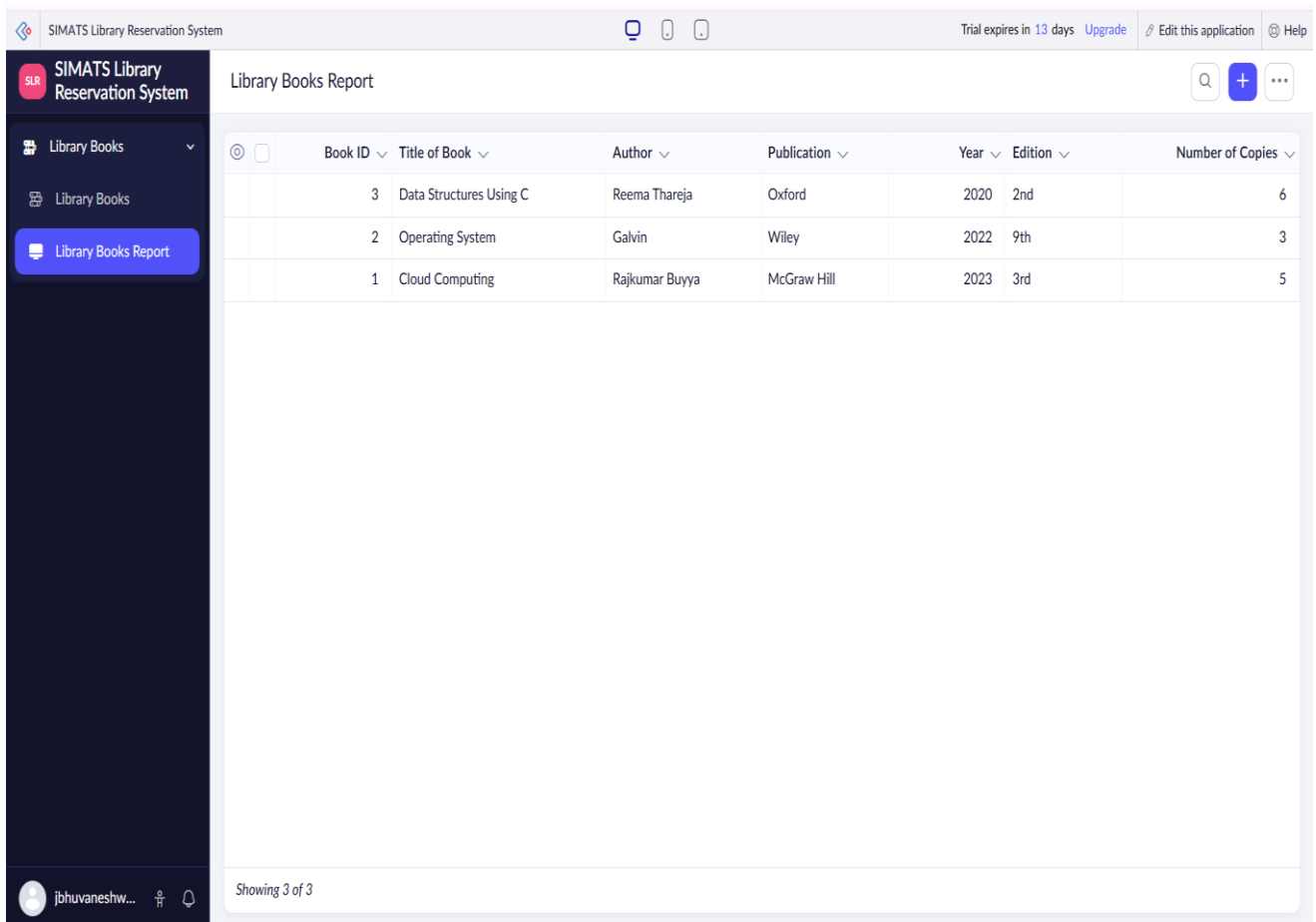
Experiment 2: Create a simple cloud software application for Property Buying & Rental process (In Chennai city) using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Buy, Rent, commercial, Rental Agreement , Rental Agreement, Area, place, Range, property value etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 5 CPU, 12 GB RAM and 8 GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

Assessment Tasks

Experiment 1: Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.



The screenshot displays the SIMATS Library Reservation System (SLR) interface. The top navigation bar includes the system name, user avatars, and trial status (13 days left). The left sidebar shows the 'Library Books Report' menu item. The main content area features a table with the following data:

Book ID	Title of Book	Author	Publication	Year	Edition	Number of Copies
3	Data Structures Using C	Reema Thareja	Oxford	2020	2nd	6
2	Operating System	Galvin	Wiley	2022	9th	3
1	Cloud Computing	Rajkumar Buyya	McGraw Hill	2023	3rd	5

The interface also includes a search bar, a '+', and a '...' menu icon in the top right. The bottom status bar shows the user 'jbhuvaneshw...' and a 'Showing 3 of 3' indicator.

Assessment Tasks

Experiment 2: Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.

Payroll Processing System
Employee Payroll

Basic Fields

Employee ID, Overall Experience, Experience In Present Org..., Employee Name, Basic Pay, HRA

Field Properties

Field name: Employee ID
Field link name: Employee_ID
Validation: Start Index: 1
Appearance: Field type: Auto Number

Payroll Processing System
Employee Payroll Report

Present Organization	Overall Experience	Basic Pay	HRA
1	3	₹ 25,000.00	₹ 5,000.00
3	7	₹ 35,000.00	₹ 7,000.00
2	5	₹ 25,000.00	₹ 5,000.00

Showing 3 of 3

Employee ID: 1
Employee Name: Rahul
Experience in Present Organization: 2
Overall Experience: 5
Basic Pay: ₹ 25,000.00
HRA: ₹ 5,000.00

Assessment Tasks

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

The screenshot displays the VMware Workstation interface. On the left, the 'Library' pane shows 'My Computer' with the 'Ubuntu 64-bit' virtual machine selected. The main window has tabs for 'Home' and 'Ubuntu 64-bit'. The 'Ubuntu 64-bit' tab is active, showing options to 'Resume this virtual machine' and 'Edit virtual machine settings'. Below these are sections for 'Devices' and 'Description'. The 'Devices' section lists various hardware components and their settings. The 'Description' section has a text input field. On the right, a large black rectangle represents the virtual machine's display. At the bottom right, the 'Virtual Machine Details' section provides summary information.

Devices	
Memory	4 GB
Processors	2
Hard Disk (SCSI)	20 GB
Hard Disk 2 (SCSI)	2 GB
CD/DVD (SATA)	Auto detect
Network Adapter	NAT
USB Controller	Present
Sound Card	Auto detect
Printer	Present
Display	Auto detect

Virtual Machine Details	
State:	Suspended
Snapshot:	vmare 1
Configuration file:	C:\Users\bhuva\OneDrive\Machines\Ubuntu 64-bit\...
Hardware compatibility:	Workstation 16.2.x virtual
Primary IP address:	Network information is not available

Assessment Tasks

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

